

Student Portal Models and API Endpoints (Recommended)

Recommended Models

1. **Student** - Student profiles
2. **StudentEnrollment** - Unit enrollments
3. **StudentDevice** - Registered BLE devices
4. **AttendanceRecord** - Attendance marking
5. **AttendanceSession** - Active sessions view
6. **BeaconLog** - Beacon detection logs

Recommended API endpoints for the Student Portal based on the models:

Authentication

- `POST /api/student/auth/login` - Student login
- `POST /api/student/auth/logout` - Student logout
- `POST /api/student/auth/change-password` - Change password
- `POST /api/student/auth/forgot-password` - Request password reset

Profile

- `GET /api/student/profile` - Get student profile
- `PUT /api/student/profile` - Update student profile
- `GET /api/student/profile/statistics` - Get attendance statistics

Enrollments

- `GET /api/student/enrollments` - Get enrolled units
- `GET /api/student/enrollments/:enrollmentId` - Get enrollment details

Timetable

- `GET /api/student/timetable` - Get student timetable
- `GET /api/student/timetable/upcoming` - Get upcoming classes

Attendance

- `GET /api/student/attendance/sessions/active` - Get active sessions
- `POST /api/student/attendance/mark` - Mark attendance

- `GET /api/student/attendance/history` - Get attendance history
- `GET /api/student/attendance/statistics` - Get attendance stats

Devices

- `POST /api/student/devices/register` - Register new device
- `GET /api/student/devices` - Get registered devices
- `PUT /api/student/devices/:deviceId/primary` - Set primary device
- `DELETE /api/student/devices/:deviceId` - Deactivate device

Lecturer Portal Models and API Endpoints (Recommended)

Recommended models

1. **Lecturer** - Lecturer profiles
2. **LecturerAssignment** - Unit assignments
3. **TimetableEntry** - Class schedules
4. **AttendanceSessionManagement** - Session CRUD
5. **AttendanceVerification** - Verify/correct attendance

Recommended API endpoints for the Lecturer Portal based on the Models:

Authentication

- `POST /api/lecturer/auth/login` - Lecturer login
- `POST /api/lecturer/auth/logout` - Lecturer logout
- `POST /api/lecturer/auth/change-password` - Change password

Profile

- `GET /api/lecturer/profile` - Get lecturer profile
- `PUT /api/lecturer/profile` - Update lecturer profile
- `GET /api/lecturer/statistics` - Get lecturer statistics

Assignments

- `GET /api/lecturer/assignments` - Get assigned units
- `GET /api/lecturer/assignments/:assignmentId` - Get assignment details

- `GET /api/lecturer/workload` - Get workload statistics

Timetable

- `GET /api/lecturer/timetable` - Get lecturer timetable
- `GET /api/lecturer/timetable/today` - Get today's classes
- `GET /api/lecturer/timetable/upcoming` - Get upcoming classes

Session Management

- `POST /api/lecturer/sessions` - Create attendance session
- `PUT /api/lecturer/sessions/:sessionId/start` - Start session
- `PUT /api/lecturer/sessions/:sessionId/end` - End session
- `PUT /api/lecturer/sessions/:sessionId/cancel` - Cancel session
- `GET /api/lecturer/sessions/:sessionId` - Get session details
- `GET /api/lecturer/sessions/active` - Get active sessions
- `GET /api/lecturer/sessions/today` - Get today's sessions
- `GET /api/lecturer/sessions/:sessionId/summary` - Get session summary

Attendance Verification

- `PUT /api/lecturer/attendance/:attendanceId/verify` - Verify single record
- `PUT /api/lecturer/sessions/:sessionId/verify-all` - Bulk verify session
- `POST /api/lecturer/attendance/manual` - Manually mark attendance
- `POST /api/lecturer/attendance/bulk` - Bulk mark attendance
- `GET /api/lecturer/attendance/unverified` - Get unverified records
- `GET /api/lecturer/sessions/:sessionId/suspicious` - Get suspicious records
- `GET /api/lecturer/sessions/:sessionId/export` - Export session attendance

Reports

- `GET /api/lecturer/reports/attendance-stats` - Get attendance statistics
- `GET /api/lecturer/reports/unit/:unitId` - Get unit attendance report
- `GET /api/lecturer/reports/student/:studentId` - Get student attendance history

Class Lists

- `GET /api/lecturer/units/:unitId/students` - Get enrolled students in a unit

Admin Portal Models and API Endpoints (Recommended)

Recommended Models

1. **Admin** - Admin users with permissions
2. **AuditLog** - System activity logging
3. **SystemSettings** - Configuration management
4. **ClassBeacon** - Beacon-classroom assignments
5. **Report** - Analytics and reporting

Based on the models, recommended API endpoints for the Admin Portal:

Authentication

- `POST /api/admin/auth/login` - Admin login
- `POST /api/admin/auth/logout` - Admin logout
- `POST /api/admin/auth/change-password` - Change password

Admin Management

- `GET /api/admin/admins` - Get all admins
- `GET /api/admin/admins/:adminId` - Get admin details
- `POST /api/admin/admins` - Create new admin
- `PUT /api/admin/admins/:adminId` - Update admin
- `DELETE /api/admin/admins/:adminId` - Delete admin
- `GET /api/admin/admins/role/:role` - Get admins by role
- `PUT /api/admin/admins/:adminId/permissions` - Update permissions

User Management

- `GET /api/admin/users` - Get all users
- `GET /api/admin/users/:userId` - Get user details
- `POST /api/admin/users` - Create new user
- `PUT /api/admin/users/:userId` - Update user
- `DELETE /api/admin/users/:userId` - Delete user
- `PUT /api/admin/users/:userId/activate` - Activate user

- `PUT /api/admin/users/:userId/deactivate` - Deactivate user

Student Management

- `GET /api/admin/students` - Get all students
- `GET /api/admin/students/:studentId` - Get student details
- `POST /api/admin/students` - Create new student
- `PUT /api/admin/students/:studentId` - Update student
- `DELETE /api/admin/students/:studentId` - Delete student

Lecturer Management

- `GET /api/admin/lecturers` - Get all lecturers
- `GET /api/admin/lecturers/:lecturerId` - Get lecturer details
- `POST /api/admin/lecturers` - Create new lecturer
- `PUT /api/admin/lecturers/:lecturerId` - Update lecturer
- `DELETE /api/admin/lecturers/:lecturerId` - Delete lecturer

Institutional Hierarchy

- `GET /api/admin/schools` - Get all schools
 - `POST /api/admin/schools` - Create school
 - `PUT /api/admin/schools/:schoolId` - Update school
 - `DELETE /api/admin/schools/:schoolId` - Delete school
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- `GET /api/admin/departments` - Get all departments
 - `POST /api/admin/departments` - Create department
 - `PUT /api/admin/departments/:departmentId` - Update department
 - `DELETE /api/admin/departments/:departmentId` - Delete department
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- `GET /api/admin/programs` - Get all programs
 - `POST /api/admin/programs` - Create program
 - `PUT /api/admin/programs/:programId` - Update program
 - `DELETE /api/admin/programs/:programId` - Delete program

- `GET /api/admin/units` - Get all units
- `POST /api/admin/units` - Create unit
- `PUT /api/admin/units/:unitId` - Update unit
- `DELETE /api/admin/units/:unitId` - Delete unit

Venue/Class Management

- `GET /api/admin/classes` - Get all classes
- `POST /api/admin/classes` - Create class
- `PUT /api/admin/classes/:classId` - Update class
- `DELETE /api/admin/classes/:classId` - Delete class

Beacon Management

- `GET /api/admin/beacons` - Get all beacons
- `POST /api/admin/beacons` - Register new beacon
- `PUT /api/admin/beacons/:beaconId` - Update beacon
- `DELETE /api/admin/beacons/:beaconId` - Delete beacon
- `GET /api/admin/beacons/unassigned` - Get unassigned beacons
- `POST /api/admin/beacons/:beaconId/assign` - Assign beacon to class
- `DELETE /api/admin/beacons/:beaconId/unassign` - Unassign beacon

Timetable Management

- `GET /api/admin/timetable` - Get all timetable entries
- `POST /api/admin/timetable` - Create timetable entry
- `PUT /api/admin/timetable/:timetableId` - Update timetable entry
- `DELETE /api/admin/timetable/:timetableId` - Delete timetable entry
- `POST /api/admin/timetable/bulk-import` - Bulk import timetable

System Settings

- `GET /api/admin/settings` - Get all settings
- `GET /api/admin/settings/:key` - Get specific setting
- `PUT /api/admin/settings/:key` - Update setting
- `POST /api/admin/settings/bulk` - Bulk update settings
- `POST /api/admin/settings/initialize` - Initialize default settings

Audit Logs

- `GET /api/admin/audit-logs` - Get audit logs
- `GET /api/admin/audit-logs/user/:userId` - Get logs for user
- `GET /api/admin/audit-logs/entity/:type/:id` - Get logs for entity
- `GET /api/admin/audit-logs/search` - Search audit logs
- `GET /api/admin/audit-logs/summary` - Get activity summary

Reports & Analytics

- `GET /api/admin/reports/attendance/overall` - Overall attendance stats
- `GET /api/admin/reports/attendance/by-program` - Attendance by program
- `GET /api/admin/reports/attendance/by-unit` - Attendance by unit
- `GET /api/admin/reports/attendance/low-students` - Low attendance students
- `GET /api/admin/reports/lecturer-performance` - Lecturer performance
- `GET /api/admin/reports/class-utilization` - Class/venue utilization
- `GET /api/admin/reports/beacon-usage` - Beacon usage statistics
- `GET /api/admin/reports/daily-trends` - Daily attendance trends
- `GET /api/admin/reports/student/:studentId` - Student detailed report
- `POST /api/admin/reports/export` - Export attendance data

Models that will be shared across the portals

1. **BaseModel** - Abstract base with CRUD operations
2. **User** - Authentication (students, lecturers, admins)
3. **School** - Schools/faculties
4. **Department** - Departments within schools
5. **Program** - Academic programs (courses)
6. **Unit** - Course units/modules
7. **Class** - Classrooms/venues
8. **Beacon** - BLE beacon devices

9. **ProgramUnit** - Program-unit relationships

NB: The documentation above is based on the database schema, this will ensure consistency across all the portals, respective portals should implement the recommended models and endpoints (they are all accommodated by the database) , this will ensure we maintain the business logic for each portal.